

Anchored multi-omics integration and knowledge-anchored residual discovery

A never-below-the-best-view integrator, and a tool to find what the textbook misses.

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Abstract

Multi-omics integration is usually sold as “more modalities → better predictions,” yet on real cancer data a strong single modality is hard to beat and naive fusion often does worse. We define an **anchored** integrator that anchors on the empirically strongest (or a fixed prior) modality and adds the others only as a non-negative gated residual, so it is provably never below its best single view and improves on it only where another modality carries orthogonal signal. Across nine TCGA-BRCA endpoints the gate adds nothing where one modality dominates and engages only on a constructed positive control — quantifying, rather than assuming, where multi-omics helps. Generalising the anchor from “best modality” to **established knowledge** (a textbook gene signature, a published clock) yields a zero-parameter prior that nearly matches a fully trained model and gives the programme’s only clean super-additive gain. Finally, gating genome-wide data on a textbook prior and mining the *residual* turns the integrator into a **discovery** tool: it recovers a verified basal/lineage axis beyond “LumA/B = proliferation” (keratinization, $p \approx 8 \times 10^{-11}$) and a verified neuroendocrine/immune axis beyond the ERBB2 amplicon, while correctly finding nothing where the textbook is already complete (ER). The method is modality-agnostic and, above all, honest: discovery where warranted, silence where not.

1. Background

The prevailing assumption that adding omics layers improves prediction is contradicted by careful benchmarks: a large TCGA survival study found that adding data types most often *hurt* performance and that mRNA ($\pm$ miRNA) sufficed for most cancers (Li et al., 2024); late-fusion work shows the dominant modality is task-specific and pan-cancer multi-omics barely exceeds the best single view (Nikolaou et al., 2023); and naive concatenation routinely underperforms structured fusion (Makrodimitis et al., 2023). The honest goal is therefore not “always fuse” but **never do worse than the best single modality, and improve on it only where another modality carries signal the leader cannot**. This note defines an integrator built to that contract and then repurposes it as a discovery engine.

2. Method

All functions live in `omniomics.multiomics`.

Anchor selection (`select_anchor`). Score each modality by repeated stratified-CV performance (primary), tie-break by robustness (subtract a multiple of the CV s.d., weight by coverage). The anchor is the top composite, chosen empirically per task inside the outer CV.

Gated residual (`anchored_gate` / `anchored_integrate`). Pin the anchor; add each secondary on the anchor’s residual with a non-negative weight β chosen on held-out data, $\beta = 0$ allowed. The combined score is $\text{logit}(\text{anchor}) + \beta \cdot \text{secondary}$. $\beta = 0 \equiv \text{combined} \equiv \text{anchor}$ (never below); $\beta > 0$ only where a secondary earns a margin gain. A constrained, interpretable special case of cooperative learning (Ding et al., 2022).

Forward gating (`forward_integrate` / `auto_integrate`). For ≥ 3 modalities, add one at a time onto the current residual in rank order, keeping each only if it clears the margin — forward selection over modalities, with an interpretable record of which entered and at what β .

Knowledge anchor (knowledge_anchored_integrate / signature_score). The anchor need not be a data modality: a fixed textbook signature or published clock (zero trained parameters) becomes the anchor and the data is gated onto its residual. The gate then answers a clinically meaningful question — *does the genome-wide data add anything beyond established biology?*

Residual discovery (anchored_residual_discovery). Anchor on the prior, gate the data, test that the residual is signal-not-noise, and surface the features driving it that are *orthogonal* to the anchor — candidate biology beyond the textbook — with a matched-random-panel null as the noise control.

Small-sample robustness throughout: a secondary is added only if a non-zero β beats the anchor by more than a gate_margin, with each β 's inner-CV score averaged over repeated splits.

3. Results (TCGA-BRCA)

Anchored integration never falls below the leader. On LumA/B, RNA alone scores 0.947 and methylation 0.745; a naive stack dips to 0.941 (below RNA) while the gated combiner stays at 0.947 ($\beta = 0$ in 9/10 repeats). A constructed positive control — a held-out methylation set RNA cannot see — flips this: the gate engages ($\beta = 4-8$) for a significant +0.047. So the gate is both protective and capable.

Not RNA-biased. Run blind on nine expert-defined subtype labels, auto_integrate anchors on methylation for all five methylation-defined clusters and on RNA for all four PAM50 calls — anchor selection follows the biology, never below the leader on any of the nine.

Knowledge anchoring — the first clean super-additive gain. A 20-gene proliferation index with zero trained parameters scores 0.919 on LumA/B (nearly the trained 1500-gene RNA model, 0.942); gating genome-wide data reaches **0.947 ($\Delta +0.029$)**, beating the pure data model. With the Horvath clock as the anchor for normal-tissue age the clock alone (0.947) already beats RNA (0.911) and data adds ≈ 0 — the textbook suffices and the gate says so.

Residual discovery, verified. Mining the proliferation-anchored residual on LumA/B surfaces the basal/ squamous-lineage axis (KRT5/14/17/6B, TP63, DSG3/DSC3, SOX10, COL17A1, KLK5/7/8; partial $r \approx 0.46$). It is verified three ways: a train-selected panel beats random panels on a held-out test in 10/10 splits (no selection leakage), the basal core recurs in 10/10 splits, and under permuted labels the advantage collapses (real +0.021 > all permutations). g:Profiler confirms a coherent program — cornified-envelope formation / keratinization / epidermis development (Reactome $p \approx 8 \times 10^{-11}$).

Generalises, reproduces, and discriminates.

endpoint	textbook anchor (0 params)	anchor AUROC	Δ data adds	verdict
LumA vs LumB	proliferation index (20 genes)	0.919	+0.029	incomplete → basal/lineage axis (verified)
HER2 status	ERBB2 amplicon (9 genes)	0.752	+0.054	incomplete → neuroendocrine/immune axis (verified, $\neq$ basal)
ER status	ER/luminal signature (20 genes)	0.938	−0.001	complete → no hidden axis (specificity)

The HER2 discovery is a second, *different* verified axis (held-out +0.068 in 8/8 splits; panel $p = 0.024$; label-specific), a diffuse neuroendocrine/secretory + immune program distinct from basal.

ER is a real-data specificity negative: the textbook signature already matches genome-wide RNA.

Modality-agnostic. The discovery runs unchanged on DNA-methylation features; on LumA/B it correctly finds no methylation axis (random and basal-targeted CpGs both $\Delta \approx 0$) — the basal axis is transcriptional, not predictively methylation-encoded here.

4. Related work

The approach instantiates three established traditions: clinical-offset / incremental-value modelling, where an established model is an offset and omics are selected on the residual (Volkman et al., 2019, a near-exact methodological twin); biologically-informed models that bake known biology into structure (Elmarakeby et al., 2021, P-NET; Liu et al., 2024, Pathformer); and ML on established signatures (Sarafidis et al., 2024). Our contribution is the packaged, **never-below-anchor, margin-gated** residual on a zero-parameter prior, plus the residual-discovery framing with a matched-random-panel noise control.

5. Limitations

Predictive gains over a strong anchor are modest; the method’s value is *routing and discovery*, not large AUROC wins. Genuine super-additive multi-omics gains are rare on real endpoints (one constructed positive control; one real knowledge-anchored gain). The discovery validations use TCGA-BRCA; a truly external cohort is the natural next test. Discovered axes are candidate hypotheses, not causal claims.

6. Conclusion

Anchor on the leading view (or the textbook), add the rest as a gated residual, and let the gate tell you — honestly — whether the genome-wide data beats established biology. Where it does, the same residual names the new axis (basal for LumA/B; neuroendocrine/immune for HER2); where it does not (ER), the method stays silent. Discovery where warranted, silence where not.

References

Li et al., *BMC Med Inform Decis Mak* 2024 · Nikolaou et al., *Cancer Res* 2023 · Makrodimitris et al., *Brief Bioinform* 2023 · Ding, Li, Narasimhan & Tibshirani, *PNAS* 2022 · Volkman et al., *BMC Med Res Methodol* 2019 · Elmarakeby et al., *Nature* 2021 · Liu et al., *Bioinformatics* 2024 · Sarafidis et al., *ESMO Open* 2024 · Morandini et al., *GeroScience* 2023 · Horvath, *Genome Biology* 2013.

Code, runners, recorded metrics and CI guards: the omniomics-prototype repository. Full method note: reports/anchored_integration_methods.md.